G NIHAL

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Computer Science and Engineering student interested in software development, focused on building practical web applications and full-stack software solutions. Hands-on experience developing full-stack projects across frontend interfaces, backend APIs, databases, and authentication. Continuously strengthening software engineering and problem-solving skills through Data Structures and Algorithms and project development.

TECHNICAL SKILLS

Languages	Java, JavaScript, Python, C#, SQL, HTML5, CSS
Frontend	React.js, Tailwind CSS, Bootstrap, Responsive Web Design
Backend	Node.js, Express.js, .NET, REST APIs, JWT Authentication
Databases	PostgreSQL, MongoDB, MySQL
CS Fundamentals	Data Structures, Algorithms, OOP, DBMS
Tools	Git, GitHub, Visual Studio Code

PROJECTS

SabAI Pay – AI-Powered UPI Payment Assistant

Tech Stack: React, Node.js, Express, Supabase PostgreSQL, Google Gemini AI, Razorpay, JWT

- Built a prototype AI-powered UPI payment assistant demonstrating Agent Pay and Reserve Pay through conversational, policy-constrained agentic payments
- Designed a separation-of-concerns architecture where Gemini proposes structured intents, a deterministic policy engine authorizes payments, and the payment engine executes auditable transactions
- Implemented AI chat flows for UPI payments, mobile recharge, bill payments, food ordering, auto-pay scheduling, and SabAI Gems cashback with Reserve Pay spending limits
- Integrated JWT authentication, mock bank/UPI flows, Razorpay test-mode payments, transaction auditing, and multi-payment method support across UPI, bank account, Reserve Pay, and SabAI Gems

Repository: github.com/nihal705/sabai-pay

MyCodeNotes – Full Stack Learning & Coding Platform

Tech Stack: React, Tailwind CSS, Python, FastAPI, PostgreSQL, SQLAlchemy, JWT, pgvector, RAG, Groq, Gemini

- Built a structured learning platform with four searchable, filterable modules — 80+ solved LeetCode problems, 250+ practice problems, 200+ concept references, and 9+ personal notes — with dual-language (Java/Python) solutions across Easy/Medium/Hard tiers
- Engineered NextraAI, a domain-specific AI assistant using retrieval-augmented generation (RAG) over 2,500+ pgvector-embedded content chunks, with tiered knowledge routing, SSE streaming, source citations, per-user rate limiting, and Groq → Gemini model fallback
- Developed a FastAPI + PostgreSQL REST backend (~50 endpoints, 17 SQLAlchemy-modeled tables) with JWT authentication, bcrypt password hashing, Pydantic validation, and email OTP signup — powering progress tracking, streaks, AI study plans, and timed mock tests
- Designed a responsive React frontend with dark/light mode, Framer Motion animations, a full-screen chat interface with voice input and multi-source context pinning, and PDF/CSV progress export, deployed on Vercel and Render with Git-based automatic deployments

Live Demo: https://mycodenotes.dev | Repository: github.com/nihal705/MyCodeNotes

InkWell – Multi-Author Article Publishing Platform

Tech Stack: .NET 10 Web API, C#, EF Core, Neon PostgreSQL, JWT, Cloudinary, Vanilla JavaScript SPA, Docker

- Built a full-stack article publishing platform where anyone can register, write, and submit articles, with admin review gating every post before it goes live
- Engineered a .NET 10 REST API with EF Core and Neon PostgreSQL covering post CRUD, search, category filtering, author profiles, admin dashboard, and a rejection-feedback loop for resubmission
- Implemented JWT auth with BCrypt hashing and role-based access, Cloudinary image uploads, and a responsive vanilla JS SPA with client-side routing and light/dark theme — deployed on Render via Dockerfile

Live Demo: https://inkwell-3m7y.onrender.com | Repository: github.com/nihal705/inkwell

EDUCATION & CERTIFICATIONS

Bachelor of Engineering (B.E.) – Computer Science & Engineering *2023 – 2027*

S.E.A College of Engineering & Technology
CGPA: 8.69

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

Certifications: React and Redux | From Relational Model (SQL) to MongoDB's Document Model | Ethical Generative AI in Software Engineering